

MALAQUITE BEACH, TX, USA

WMO: 997366

Lat: 27.581N

Lon: 97.216W

Elev: 10

StdP: 101.20

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 2.9	(c) 5.2	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 14.7	(k) 12.3	(l) 13.1	(m) 12.0	(n) 8.1	(o) 350	(p) 0.609

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 2.0	(c) 31.7	(d) N/A	(e) 30.5	(f) N/A	(g) 29.8	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 5.7	(p) 100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 13.1	(b) 12.0	(c) 11.1	(e) 1.6	(f) 33.0	(g) 2.2	(h) 2.3	(i) 0.0	(j) 34.6	(k) -1.2	(l) 36.0	(m) -2.5	(n) 37.3	(o) -4.0	(p) 38.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	22.1	13.6	15.3	17.8	21.8	24.8	27.6	27.9	28.4	27.6	24.7	19.7	15.7		
	DBStd	6.03	3.93	4.79	3.40	2.42	1.81	1.04	1.08	0.99	1.61	3.00	3.93	4.00		
	HDD10.0	37	16	11	3	0	0	0	0	0	0	0	2	6		
	HDD18.3	436	150	99	48	6	1	0	0	0	0	2	32	99		
	CDD10.0	4455	126	159	243	353	461	528	556	570	528	457	292	182		
	CDD18.3	1812	2	14	31	108	202	278	297	312	278	201	72	17		
	CDH23.3	17547	1	149	6	211	1417	3133	3514	3839	3292	1799	183	4		
	CDH26.7	4843	0	79	1	14	72	802	1086	1388	1113	287	2	0		
(14) Wind	WSAvg	6.4	5.9	6.1	6.1	7.0	7.4	7.0	6.5	5.7	6.2	6.5	6.1	5.9		
(15) Precipitation	PrecAvg	829	41	44	59	54	90	68	62	55	135	94	56	40		
	PrecMax	1222	141	120	126	232	312	221	352	171	377	417	181	177		
	PrecMin	455	0	2	2	5	2	0	0	3	24	0	1	6		
	PrecStd	217	37	37	41	53	81	54	84	51	80	95	51	43		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.2	33.2	23.9	28.8	29.1	32.0	32.3	32.4	32.3	30.5	26.3	23.5		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	20.0	30.0	22.8	25.7	28.0	30.2	30.5	31.1	30.7	29.2	25.6	22.4		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	19.2	20.1	21.9	25.0	27.4	29.3	29.7	30.2	30.0	28.7	25.2	21.8		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	DB	18.6	19.4	21.3	24.5	26.9	28.9	29.3	29.7	29.5	28.2	24.6	20.9		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.8	4.3	3.2	2.8	2.0	1.8	1.9	2.0	3.0	3.6	4.5	5.1		
		MCDBR	5.2	4.8	3.5	4.3	2.9	3.2	2.6	3.1	3.7	3.2	4.4	5.1		
	5% WB	MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.364	0.370	0.389	0.432	0.470	0.447	0.464	0.451	0.440	0.394	0.379	0.355		
	taud		2.479	2.463	2.406	2.290	2.215	2.344	2.314	2.349	2.342	2.467	2.463	2.517		
	Ebn at Noon		879	903	903	869	831	845	831	841	842	866	855	872		
	Edh at Noon		95	104	116	134	145	127	130	125	123	102	96	88		
(44) All-Sky Solar Radiation	RadAvg		3.01	3.59	4.58	5.46	6.12	6.72	6.62	6.33	5.22	4.57	3.42	2.70		
	RadStd		0.33	0.59	0.50	0.52	0.41	0.33	0.52	0.34	0.43	0.46	0.35	0.36		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (27 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+116	+103

Nomenclature: See separate page